

Question.1

Tableau is a powerful data visualization and business intelligence (BI) tool used to analyze, visualize, and share data through interactive dashboards and reports. It allows users to connect to multiple data sources, perform real-time analysis, and create visualizations without requiring extensive programming knowledge. Tableau helps organizations identify trends, patterns, and insights quickly, enabling better decision-making.

Tableau is preferred over traditional spreadsheet tools such as Microsoft Excel because it can efficiently handle large volumes of data, supports interactive dashboards, and offers advanced visualization features like maps, heat maps, treemaps, and drill-down analysis. It also enables live connections to databases, automatic data refresh, and easy sharing of dashboards, making data analysis faster, more accurate, and more collaborative than spreadsheets.

Industries where Tableau is commonly used

1. Healthcare

Hospitals and healthcare organizations use Tableau to monitor patient outcomes, track disease trends, analyze hospital performance, and optimize resource allocation.

2. Retail

Retail companies use Tableau to analyze sales performance, customer purchasing behavior, inventory levels, and regional demand to improve business decisions.

Example Use Case (Retail)

A retail company uses Tableau to create an interactive sales dashboard that displays revenue by region, product category, and month. Managers can apply filters, drill down into specific products or locations, identify top-selling products, detect low-performing regions, and make data-driven decisions to increase sales and improve inventory management.

Question.2

a) The Sample – Superstore dataset contains the following main types of fields:

1. Time-based Fields (Date Fields) - Used to analyze data over time.

Examples: Order Date, Ship Date

2. Geographical Fields - Used to create maps and perform location-based analysis. Examples: Country/Region, State/Province, City, Postal Code, Region

3. Categorical (Text/Dimension) Fields - Used to categorize and group data. Example: Category, Sub-Category, Customer Name, Customer ID, Product Name, Segment, Ship Mode

4.Numerical (Measure) Fields - Used for calculations and aggregations. Examples: Sales, Profit,Quantity, Discount

b) Identify any four Dimensions and any four Measures

Dimensions

1. Category
2. Customer Name
3. Product Name
4. Region

Measures

1. Sales
2. Profit
3. Quantity
4. Discount

These dimensions are used to categorize data, while the measures are used to perform calculations such as SUM, AVG, COUNT, MIN, and MAX for analysis and visualization in Tableau.

Question.3

1. Which field goes to Rows?

> Sales

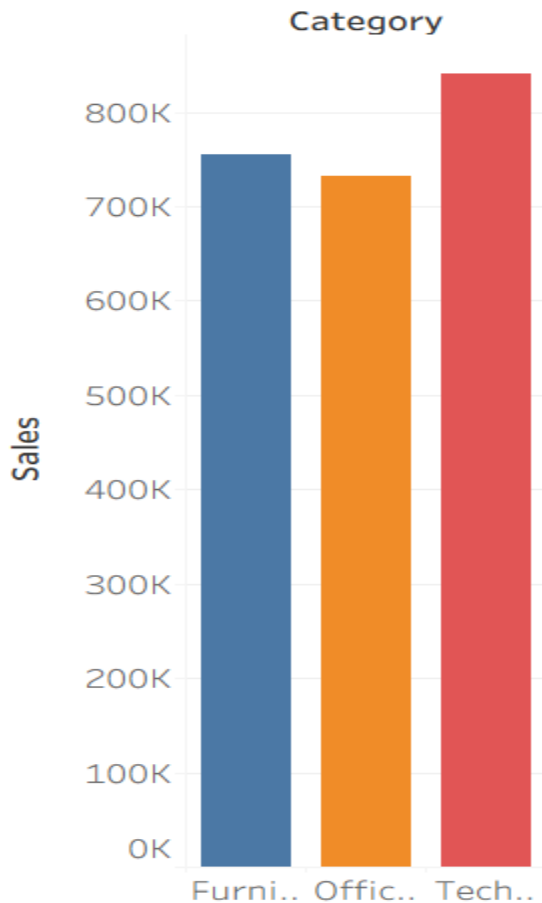
2. Which field goes to Columns?

> Category

3. Which aggregation is applied to Sales?

> SUM (Sum of Sales)

Tableau automatically aggregates the Sales measure using SUM, so the chart displays the total sales for each product category.

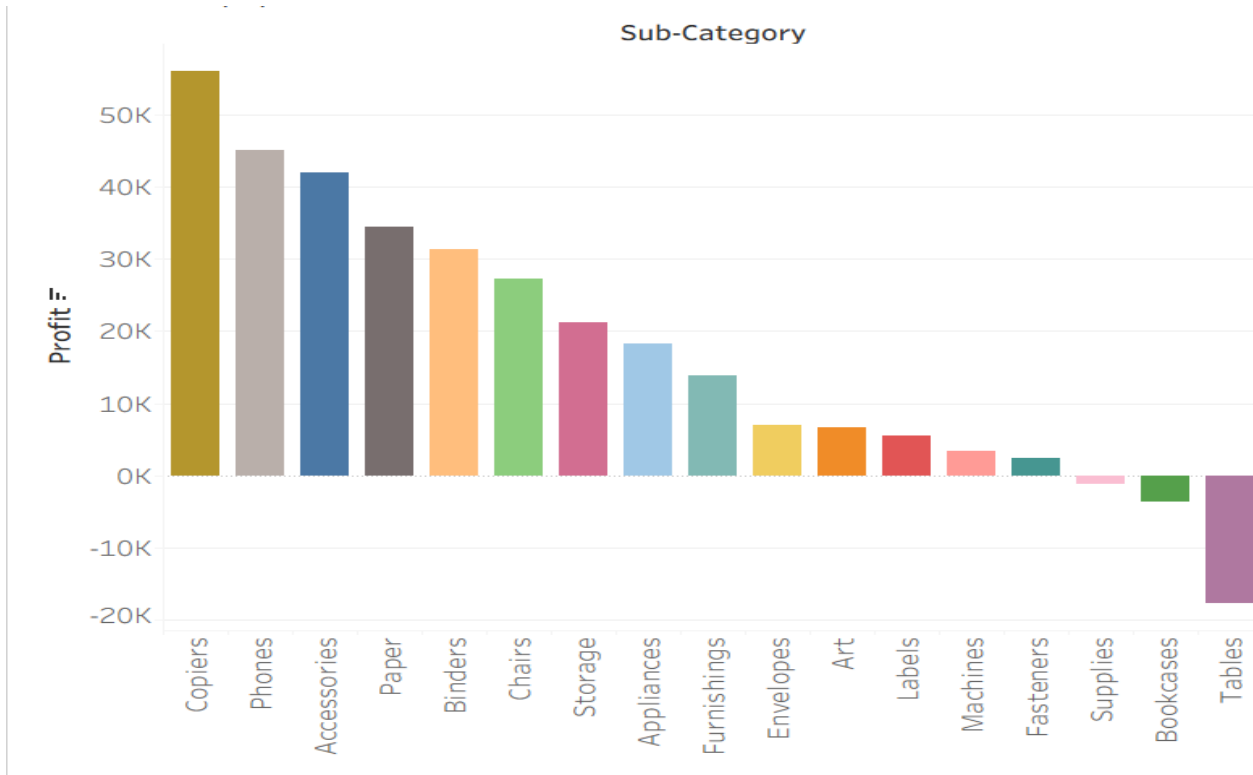


Question.4

Comparing Sales and Profit helps determine whether high sales are actually generating good earnings. A product or category may have high sales but low or even negative profit due to high discounts, production costs, or shipping expenses. This comparison helps businesses identify profitable products, reduce losses, and make better pricing and inventory decisions.

Field Placement

- > Rows: Sub-Category
- > Columns: Profit
- > Aggregation Applied: SUM(Profit)
- > Chart type: Bar Chart



Question.5

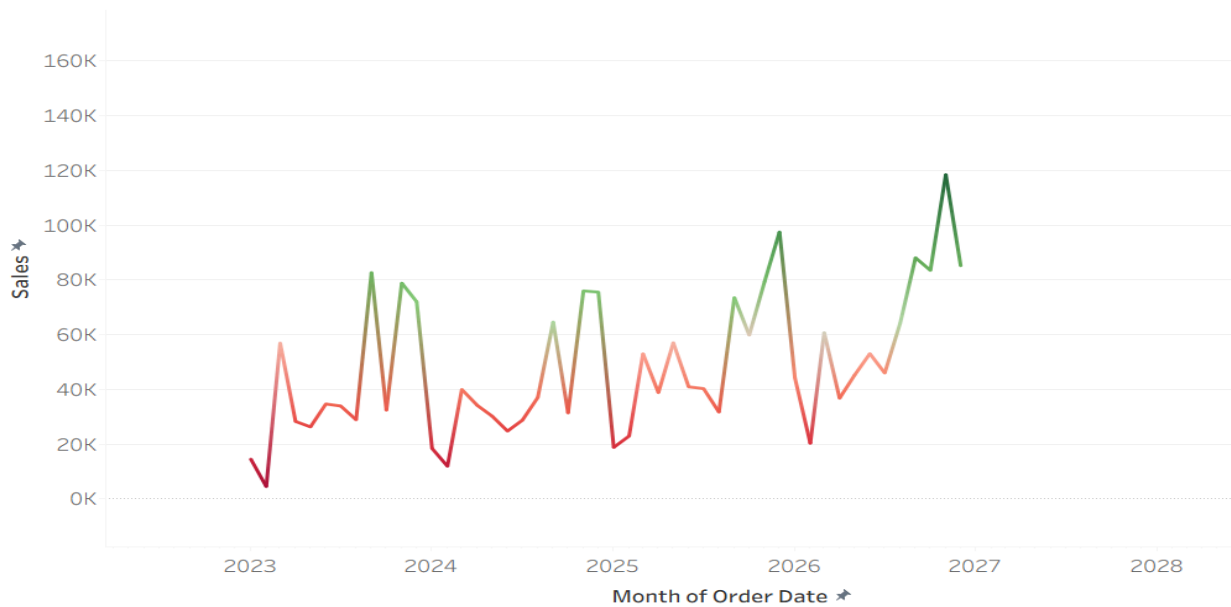
Month is the best choice.

Reason:

- It provides a detailed view of sales performance over time.
- Monthly data helps identify seasonal trends, peaks, and declines more effectively than yearly or quarterly data.
- It offers enough detail without making the chart too complex.

Line charts are ideal for time-series analysis because they connect data points in chronological order, making it easy to:

- Identify trends and patterns over time.
- Observe increases or decreases in sales.
- Detect seasonal fluctuations and recurring patterns.
- Compare performance across different time periods.
- Forecast future sales based on historical trends.



Question.6

Sub-Catego..	Category	Region
Accessories	Technology	West
Copiers	Technology	West
Machines	Technology	West
Phones	Technology	West

Steps to Apply the Filter

1. Open the worksheet containing your visualization.
2. Drag the Region field from the Data Pane to the Filters shelf.
3. In the Filter dialog box, select West and click OK.
4. Drag the Category field to the Filters shelf.
5. In the Filter dialog box, select Technology and click OK.
6. The visualization will now display only the data for the West region and Technology category.

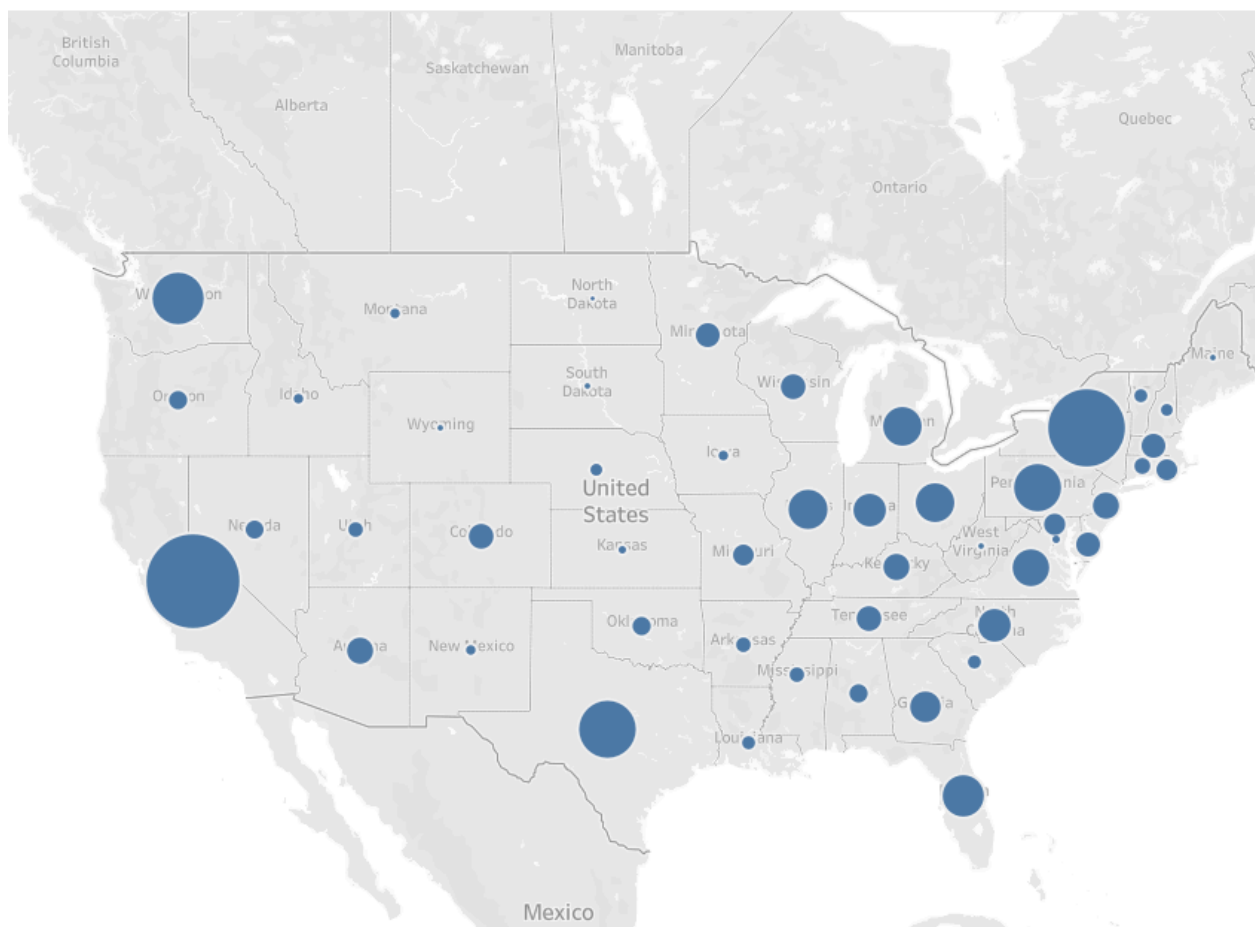
The Filters Shelf is used to apply filters in Tableau.

Question.7

Geographical analysis helps businesses understand how sales and performance vary across different locations such as states, cities, or regions. It enables organizations to identify high-performing and low-performing areas, understand customer demand, optimize inventory and distribution, plan marketing campaigns, and make informed decisions about expansion and resource allocation.

Steps to Create a Map Showing Sales by State

1. Open a new worksheet in Tableau.
2. Double-click the State field (or drag State to the canvas). Tableau automatically recognizes it as a geographic field and generates a map.
3. Drag Sales to the Color card on the Marks shelf.
4. (Optional) Drag Sales to Label on the Marks shelf to display sales values on the map.
5. Tableau will create a filled or symbol map where each state is displayed and colored according to its total sales



Question.8

Worksheet

A Worksheet is a single workspace used to create one visualization, such as a bar chart, line chart, or map.

Dashboard

A Dashboard is a collection of multiple worksheets and visualizations displayed together on one screen for interactive analysis.

Difference

- Worksheet: Displays one chart.
- Dashboard: Displays multiple charts.
- Worksheet: Focuses on one analysis.
- Dashboard: Provides an overall business view.

Why Dashboards are Useful

- Combines multiple visualizations in one place.
- Supports interactive filtering and comparison.
- Provides quick business insights.
- Helps decision-makers make faster, data-driven decisions.

Question.9

Tableau Prep is a data preparation tool used to clean, combine, and organize data before it is analyzed in Tableau. It provides a visual and user-friendly interface for preparing data, ensuring that the dataset is accurate, consistent, and ready for reporting and visualization. Tableau Prep simplifies data cleaning and transformation, making datasets more accurate and reliable for creating meaningful visualizations and dashboards in Tableau.

Importance in Real-World Data Projects

- Improves data quality by cleaning inaccurate data.
- Removes duplicate and unnecessary records.
- Combines data from multiple sources.
- Saves time through automated data preparation.
- Ensures reliable and accurate analysis.

Steps to Remove Null Values

1. Open the dataset in Tableau Prep.
2. Identify columns containing Null values.
3. Click on the field profile to view null records.
4. Select the Null values.
5. Choose Filter to remove them, or replace them with appropriate values if required.
6. Verify that the null values have been removed.

Steps to Remove Duplicate Records

1. Load the dataset into Tableau Prep.
2. Review the records to identify duplicate rows.
3. Use the Clean step to inspect duplicate values.
4. Remove duplicate records by keeping only unique entries.
5. Validate the dataset to ensure duplicates have been eliminated.

Steps to Change Incorrect Data Types

1. Open the dataset in Tableau Prep.
2. Locate the column with the incorrect data type.
3. Click the data type icon beside the field name.
4. Select the correct data type (e.g., String, Number, Date, Date & Time, or Boolean).
5. Verify that the values are displayed correctly after the conversion.
6. Save the cleaned dataset for further analysis.

Question.10

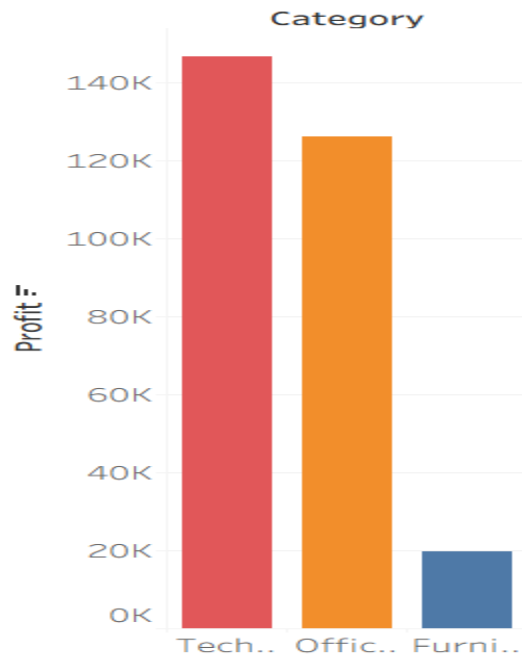
Tableau helps managers analyze business performance through interactive visualizations. By selecting appropriate charts, organizations can identify profitable categories, underperforming regions, and sales trends over time, enabling data-driven decision-making.

a) Charts to Use

1. Which category generates the highest profit?

Chart: Bar Chart

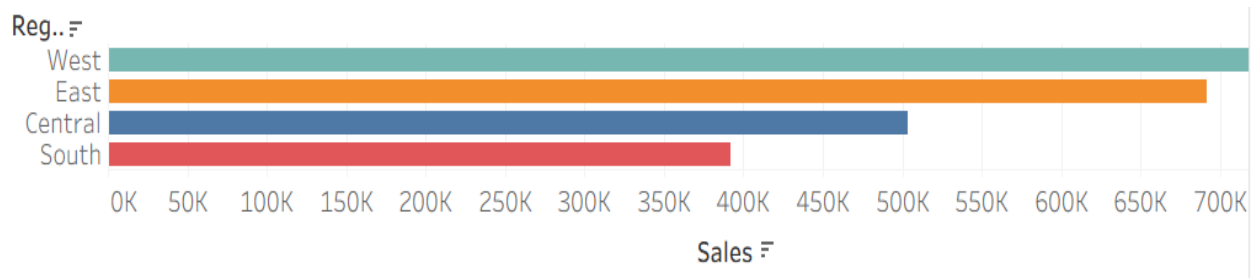
- Dimension: Category
- Measure: Profit (SUM)
- Purpose: Compares the total profit generated by each product category and identifies the most profitable category.



2. Which region performs poorly?

Chart: Bar Chart (or Map)

- Dimension: Region
- Measure: Profit or Sales (SUM)
- Purpose: Compares the performance of different regions and highlights regions with the lowest sales or profit.

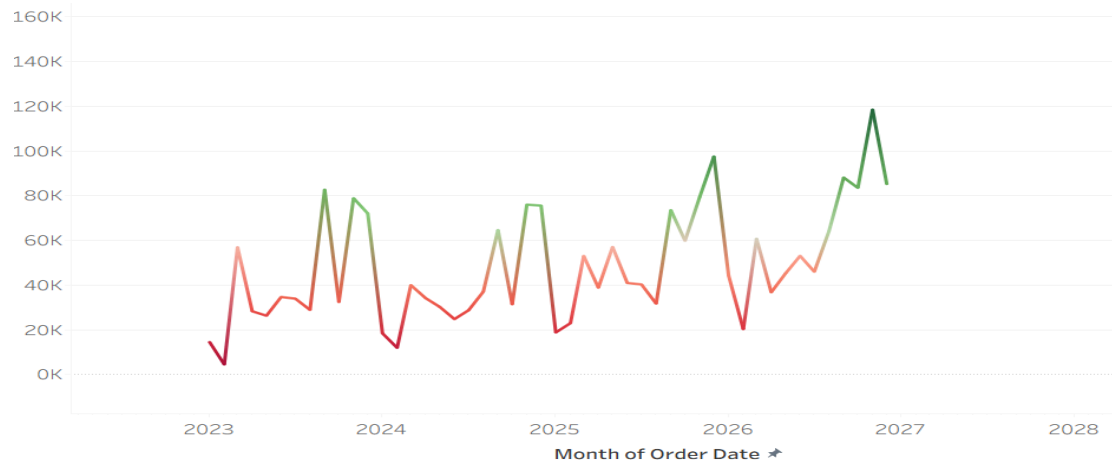


3. How does sales change over time?

Chart: Line Chart

- Dimension: Order Date (Month)

- Measure: Sales (SUM)
- Purpose: Displays sales trends over time and helps identify growth, decline, and seasonal patterns.



b) Key Insight for Management

By analyzing these charts, management can identify:

- The most profitable product category to focus on.
- The lowest-performing region that requires improvement.
- Sales trends over time, including seasonal peaks and declines.

c) Business Decision-Making

These insights help management:

- Increase investment in high-profit product categories.
- Develop strategies to improve sales in underperforming regions.
- Plan inventory and marketing campaigns based on sales trends.
- Make informed, data-driven decisions to improve profitability and business performance.

